

GED Math Unit: Data, Statistics, and Probability

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Lesson Overview

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Understanding Data

- Data: information collected from observations or measurements.
- Types:
 - Qualitative (categorical)
 - Quantitative (numerical)
- Quantitative data can be:
 - Discrete (countable values)
 - Continuous (measured values)

Visual Representations

- Bar Graphs – Compare categories.
- Histograms – Show frequency of numerical data.
- Line Graphs – Show changes over time.
- Pie Charts – Show proportions.

Example 1: Bar Graph

A survey found how many students prefer each subject:
Math: 12, Science: 9, English: 15, History: 6.

Task: Draw a bar graph.

Example 2: Histogram

Data: Test scores – 45, 55, 60, 65, 70, 70, 75, 80, 85.

Task: Create frequency intervals of 10.

Interval	Frequency
40-49	1
50-59	1
60-69	2
70-79	3
80-89	2

Example 3: Line Graph

A student's GPA over 5 semesters: 2.8, 3.0, 3.2, 3.5, 3.6.

Task: Plot points and connect them with a line.

Example 4: Pie Chart

Total students: 30

Math: 12, Science: 9, English: 9.

Angles:

$$\text{Math: } \frac{12}{30} \times 360 = 144^\circ, \text{ Science: } 108^\circ, \text{ English: } 108^\circ$$

Example 5: Stem-and-Leaf Plot

Data: 42, 45, 46, 51, 53, 53, 56, 61.

Stem — Leaf

4 — 2 5 6

5 — 1 3 3 6

6 — 1

Definitions

- Mean: Average value
- Median: Middle value when ordered
- Mode: Most frequent value
- Range: Difference between max and min

Example 1: Mean

Data: 5, 7, 9, 10, 15

$$\text{Mean} = \frac{5 + 7 + 9 + 10 + 15}{5} = 9.2$$

Example 2: Median

Data: 3, 5, 7, 9, 11, 13

Median = average of 7 and 9 = 8

Example 3: Mode

Data: 1, 2, 2, 3, 4, 4, 4, 5

Mode = 4 (appears most often)

Example 4: Range

Data: 12, 15, 17, 19, 25

$$\text{Range} = 25 - 12 = 13$$

Example 5: Combined Measures

Data: 8, 9, 9, 10, 11, 12

Mean = 9.83, Median = 9.5, Mode = 9, Range = 4.

Understanding Probability

- Probability measures how likely an event is.
- Formula:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$$

- Range: 0 (impossible) to 1 (certain)

Example 1: Coin Toss

$$P(\text{Heads}) = \frac{1}{2}$$

Example 2: Rolling a Die

$$P(\text{rolling } 4) = \frac{1}{6}$$

Example 3: Drawing Cards

Deck of 52 cards.

$$P(\text{drawing a heart}) = \frac{13}{52} = \frac{1}{4}$$

Example 4: Complementary Events

$$P(\text{not rolling } 6) = 1 - P(\text{rolling } 6) = 1 - \frac{1}{6} = \frac{5}{6}$$

Example 5: Multiple Events

$P(A \text{ and } B) = P(A) \times P(B)$ if independent.

Example: Tossing 2 heads $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Dependent Events

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

Example: Drawing 2 red cards without replacement.

$$P(\text{1st red}) = \frac{26}{52}, P(\text{2nd red}) = \frac{25}{51}$$

$$\text{So, } P = \frac{26}{52} \times \frac{25}{51} = \frac{25}{102}$$

Example 1: OR Events

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Example: Rolling a 2 or even number.

$$P(2) = \frac{1}{6}, P(\text{even}) = \frac{3}{6}, P(\text{both}) = \frac{1}{6}$$

$$P = \frac{3}{6}$$

Example 2: Conditional Probability

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

See these examples:

<https://probability.oer.math.uconn.edu/wp-content/uploads/sites/2187/2018/01/prob3160ch4.pdf>

Example 3: Tree Diagrams

- Helps us to visualize multiple outcomes.
- Usually drawn from left to right and shows ALL possible outcomes.
- Can be used for counting outcomes to find probability.

Example 4: Expected Value

If probabilities and outcomes are known:

$$E = \sum x_i p_i$$

Example: The expected value is the average outcome of a random event over many trials, calculated by multiplying the value of each outcome by its probability and summing the results. For example, a raffle ticket that costs \$3 and has a $1/150$ chance of winning a \$300 prize has an expected value of -\$1.

This is because the expected value is

$$(297 * \frac{1}{150}) + (-3 * \frac{149}{150}), \text{ which equals approximately } -\$1.$$

Example 5: Permutations/Combinations

Permutations: order matters. $nP_r = \frac{n!}{(n-r)!}$

Combinations: order doesn't matter. $nC_r = \frac{n!}{r!(n-r)!}$

Combination Example: Choose 3 students from 5: $5C_3 = 10$

A simple permutation example is arranging the letters 'a', 'b', and 'c', where the different orders like abc, acb, bac, bca, cab, and cba are all distinct permutations.

The letters a, b, and c can be arranged in 6 different ways: abc acb bac bca cab cba

Trends and Outliers

- Outlier: data point far from others.
- Trends: upward/downward patterns.
- Use scatter plots and best-fit lines.

Example 1: Scatter Plot

Height vs. Weight: correlation = positive.

Example 2: Line of Best Fit

Predict value from trend.

Equation: $y = mx + b$

Example 3: Misleading Graphs

Check for:

- Truncated axes
- Distorted scales
- Selective data

Example 4: Two-Way Tables

	Male	Female	Total
Passed	15	20	35
Failed	5	10	15
Total	20	30	50

$$P(\text{Female} \mid \text{Passed}) = \frac{20}{35} = 0.57$$

Example 5: Interpreting Mean and Spread

Standard Deviation: measures spread.

Low SD: data clustered, High SD: data spread out.

Example: Test A SD=2, Test B SD=10 → Test B more variable.

Summary

- Representing data visually
- Central tendency and spread
- Probability and combinations
- Interpreting statistical results